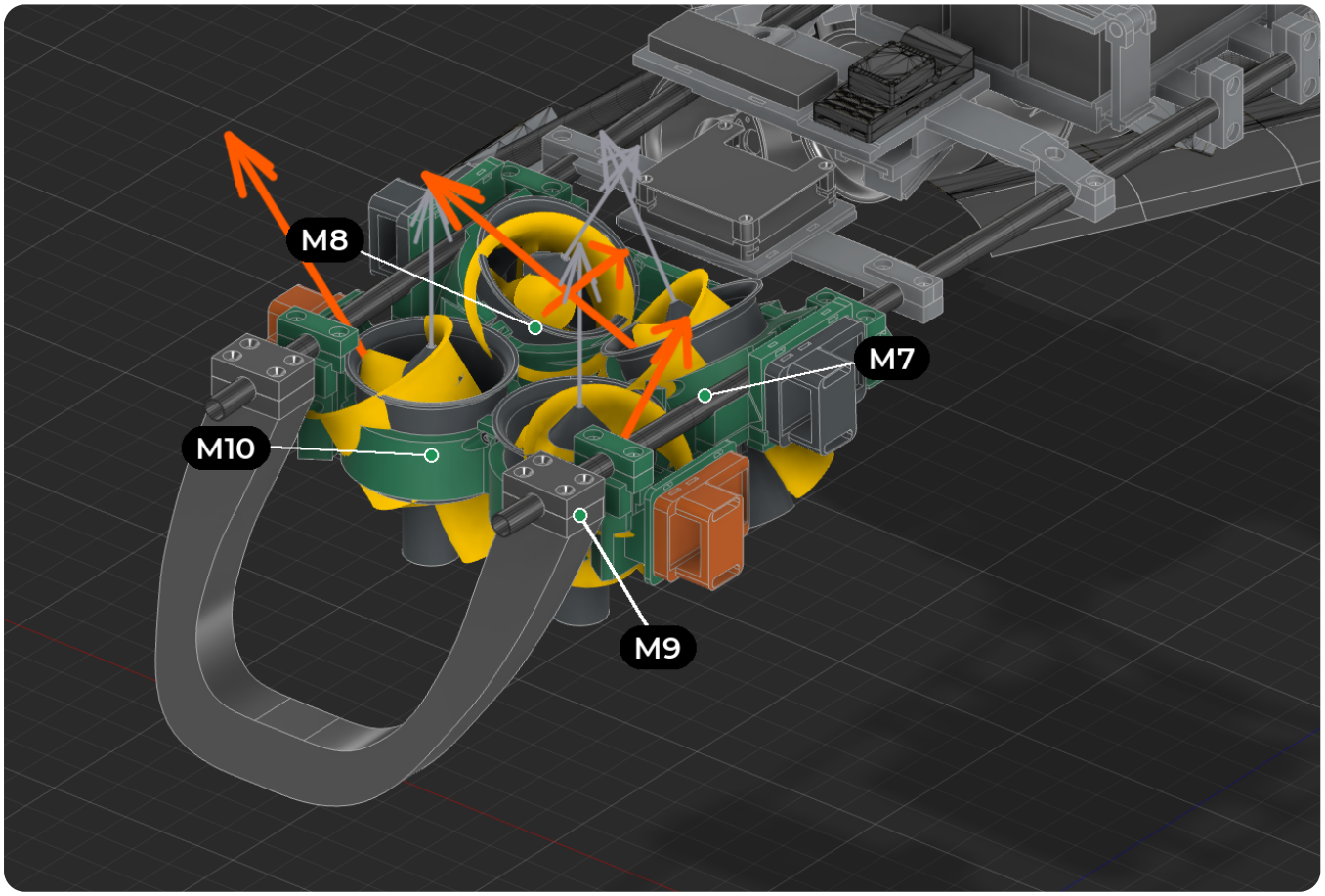
OPTION C FRONT MOTOR SETUP

ATLAS V4 with four front motors · Fusion: SMALL_SCALE_V4.2, Four front motors, Side by side (Side ESCs) · 23 October 2026



thrust axis today thrust axis, option C fan turned to option C about its own centre

CLASH CHECK IN FUSION

The fans turned to option C were intersected with the model as temporary bodies. The M7 and M8 inlets touch (0.05 cm³) and clear once each moves 2 mm outboard. No fan reaches the ESCs, the foot or the other solid parts. Each fan cuts 5.5 to 7 cm³ into today's cradle, so its seats are redrawn (page 2). Mesh parts such as the rods were not checked.

AUTHORITY AT THE HOVER, OPTION C

8.84
Yaw, rad/s²
from 2.42 as drawn

11.96
Roll, rad/s²
from 6.88 as drawn

18.55
Pitch, rad/s²
from 19.32 as drawn

14.63
Mass, kg
+1.23 on V4 today

FRONT MOTOR ANGLES

Motor	Today	Option C
M7, M8 (outer pair)	upright, 30° in	**22.6° forward, 45° in**
M9, M10 (side by side)	vertical	**11.1° forward, 30° out**

Tilt is forward from vertical; the lean (cant) is sideways, in toward the centreline or out away from it. The outer pair leans in harder and tips forward, the vertical pair leans out. Hover pitch becomes 23.8° (as drawn 17.5°).

WHY OPTION C

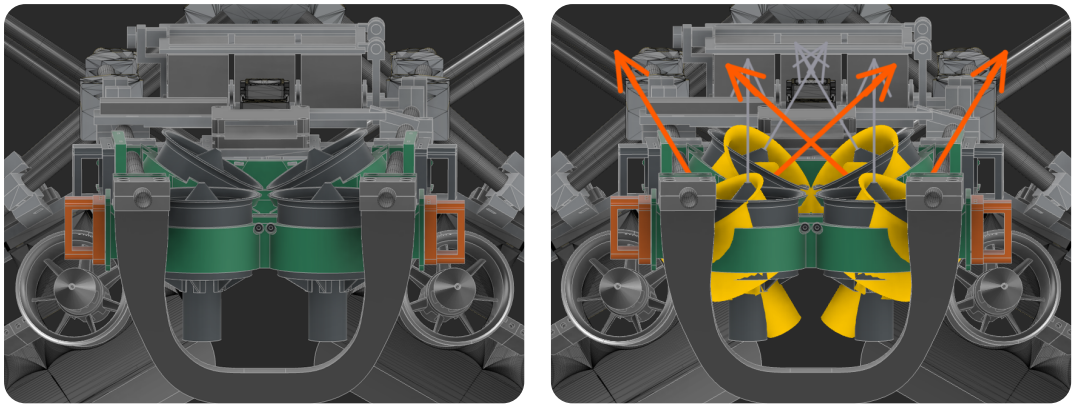
Yaw is the weakest axis on every V4 layout. Canting the front jets turns their thrust into a yaw lever 0.43 to 0.53 m ahead of the CG. Option C is the best weakest-axis result of a Monte Carlo over all four front angles (18,000 random designs, then constrained refinements) with pitch kept at 17 rad/s² or more, every angle within 45° and the side-by-side pair within 30°. With ±2° mounting error, CG ±10 mm and thrust ±5 %, the worst 5 % still give roll 10.4, pitch 16.4, yaw 7.8.

CAD CHANGES FROM TODAY'S DRAWING

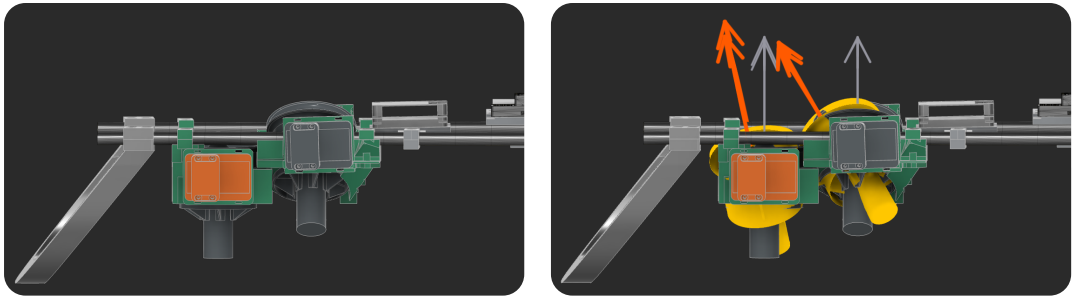
Fusion, direct modelling: Modify › Move/Copy, Rotate. Axes: X to the aircraft's right, Y up, Z aft (the nose is toward −Z). Millimetres.

TODAY

OPTION C



Nose-on, looking aft: the leans. Orange arrows and yellow fans are option C.



From the left, nose to the left: the forward tips.

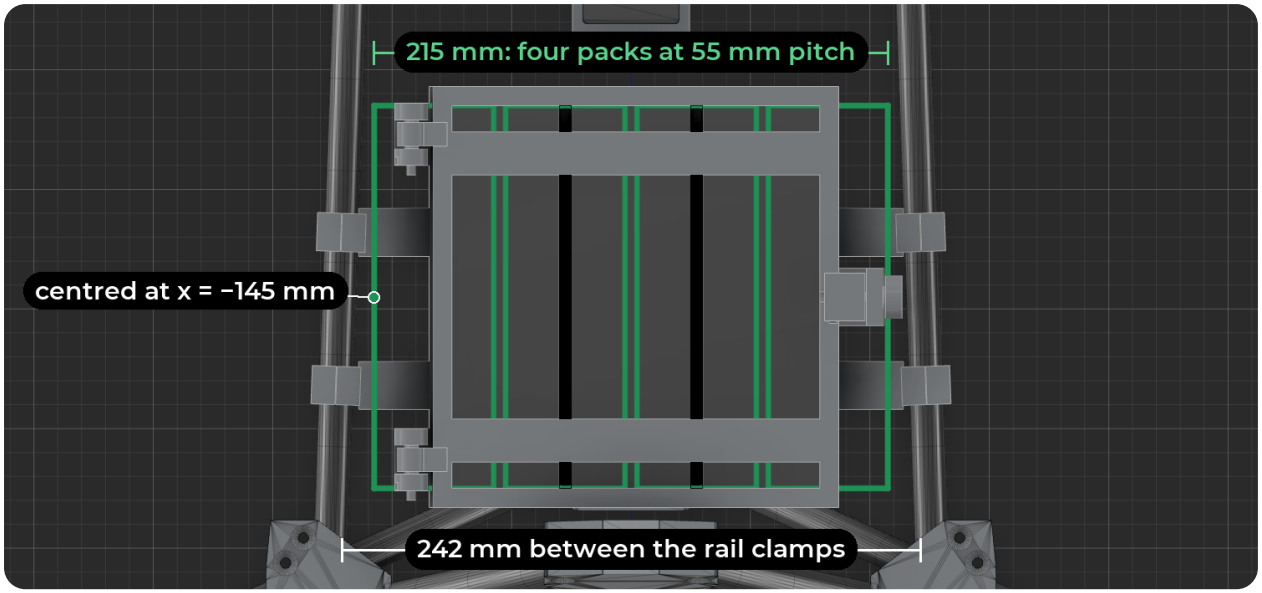
1. Work in the visible variant, **Four front motors - Side by side - Side ESCs**. Select only the fan occurrence named in the table, not its M4 lock bolts: they move with the new cradle.
2. **Rotation 1, the lean.** Axis through the fan centre, parallel to Z. Angle from the table; positive turns X toward Y (right-hand about +Z). If the axis you pick points to −Z, flip the sign.
3. **Rotation 2, the forward tip.** Axis through the same centre, in a plane parallel to X-Y, square to the fan's axis after step 2 (direction in the table). Positive tips the inlet toward the nose.
4. **Shift M7 2 mm toward −X and M8 2 mm toward +X.** Without it their inlets touch.
5. **Check every fan** against the last three columns before moving on. ±2° is enough: with CG and thrust scatter added, the worst 5 % of builds lose about 12 %.
6. **Redraw the cradle seats** (FOUR_FRONT_MOTOR_CRADLE) and the lock-bolt positions around the moved fans; today's seats overlap each turned fan by 5.5 to 7 cm^3.
7. **In AIRFRAME_DESIGNER**, fly results/v4_front4/airframe_front4_C_buildable.json. Its hover pitch is 23.8°, so the board's pitch offset (SENS_BOARD_Y_OFF) and the H-flow lens tilt follow it.

Motor	Fusion occurrence	Fan centre X, Y, Z	Rot. 1 about +Z	Rot. 2 axis X, Y, Z	Rot. 2	Shift X	From vertical	Nose-on lean	Side view
M7	MOTOR_L_REFERENCE (1)	−59.2, 512.4, −255.0	−15.0°	−0.707, 0.707, 0	+22.6°	−2 mm	49.2° (today 30°)	45.0° in	30.5° forward
M8	MOTOR_R_REFERENCE (1)	59.2, 512.4, −255.0	+15.0°	−0.707, −0.707, 0	+22.6°	+2 mm	49.2° (today 30°)	45.0° in	30.5° forward
M9	MOTOR_V_LEFT_REFERENCE (1)	−52.0, 479.5, −355.0	+30.0°	−0.866, −0.500, 0	+11.1°	none	31.8° (today 0°)	30.0° out	12.8° forward
M10	MOTOR_V_RIGHT_REFERENCE (1)	52.0, 479.5, −355.0	−30.0°	−0.866, 0.500, 0	+11.1°	none	31.8° (today 0°)	30.0° out	12.8° forward

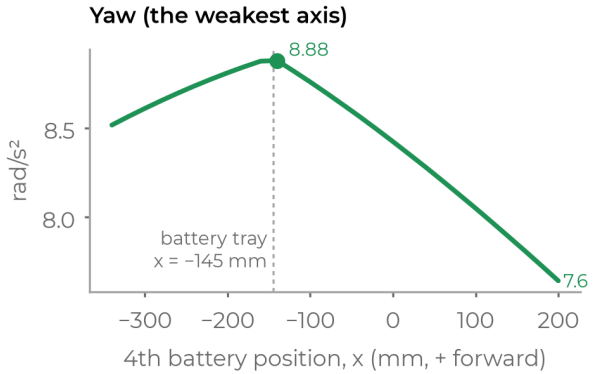
4TH BATTERY PLACEMENT

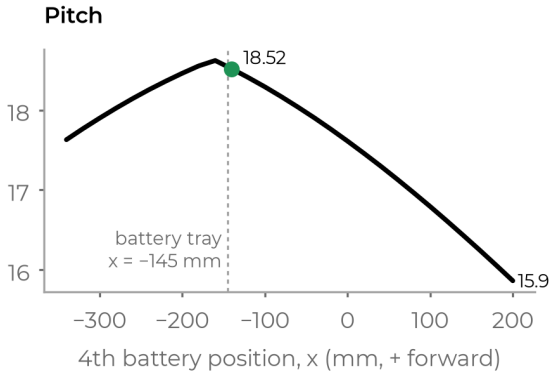
Same 700 g pack as the three (160 × 50 × 50 mm). Positions in the structural frame: x forward, y right, z down, from the reference point.

PROPOSED: FOUR PACKS SIDE BY SIDE, IN LINE WITH THE THREE (GREEN), SEEN FROM ABOVE



AUTHORITY AGAINST WHERE THE 4TH PACK SITS ON THE CENTRELINE (FRONT MOTORS AT OPTION C)





In the tray, beside the three

The authority peaks with the fourth pack at the tray (x −145 mm). Each 10 cm forward costs 0.3 to 0.4 rad/s² of yaw and 0.6 to 0.9 of pitch; on the PDB tray yaw drops to 7.9 and pitch to 16.4.

Layout	four packs, 55 mm pitch as today, centred on the centreline at x −145 mm, z −580 mm
Tray	a four-pack tray: 215 mm of packs against 160 mm today; the rail clamps leave 242 mm, so about 13 mm a side for walls
Lid	move the hinge (today y −92 mm) and the latch (y +85 to +108 mm) to the front and rear ends
Result	14.63 kg, CG x −174.5, y 1.2, z −440.7 mm
Not counted	the wider tray and lid, the 4th pack's lead

OTHER PLACES, FRONT MOTORS AT OPTION C

Placement	CG x, y, z mm	Roll	Pitch	Yaw
Four-pack tray (recommended)	−174, 1.2, −441	11.96	18.54	8.84
Stacked on the lid	−174, 1.2, −444	11.93	18.38	8.86
Forward, x 0	−168, 1.2, −441	12.17	17.61	8.42
On the PDB tray, x 150	−160, 1.2, −441	12.30	16.35	7.87
Aft of the tray, x −310	−182, 1.2, −441	11.88	17.83	8.59
Beside the three, one side	−174, 6.4, −441	12.30	19.03	9.02

IF THE TRAY CANNOT GROW

Stacked on the lid gives the same authority with the CG 3 mm higher, but the lid then cannot open without removing the pack. One pack beside the three on one side shows no loss in this static model, but it moves the CG 6 mm to the right: the aircraft hovers on uneven motors and rests unevenly on its legs. Keep it as a last resort.